

AUTONOMOUS AI SYSTEM

Continuous Executive Agent

V3.0 — The Second Brain

Memory for a 24/7 Autonomous Coding Agent



Agentic Memory

Natural-language recall



NL Retrieval

Context-aware search



Exec-Only Arch

Workers never touch mem0

What We'll Cover

01

Why Memory Matters

The problem with scattered traces and no recall

02

What We Built

The second brain architecture, taxonomy, and hooks

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How We Did It

The 8-week build journey from vision to go-live

04

What's Next

Roadmap forward from V3.0

01

Why Memory Matters

Before V3.0, the agent had traces but no memory. Every run started from zero.

Scattered Knowledge, No Recall

The agent's operational history lived in **scattered JSONL ledgers, markdown retros, and ad-hoc logs** — useful traces, but not a memory system any agent could trust or query.

Fragmented Context

Historical context spread across JSONL, markdown, and ad-hoc logs — no unified access pattern.

Unstructured Retros

Retrospective knowledge was valuable but not consistently structured for reuse across runs.

No Memory Contract

The harness lacked a stable contract for "what this system can do" and "what happened before."

Blind Worker Spawns

Every worker spawn started with zero context — no prior outcomes, no lessons learned injected.

 **Result:** Limited planning quality, repeatability, and autonomous decision-making. The agent could **execute** but not **learn**.

02

What We Built

The Second Brain: a cloud-hosted, searchable knowledge store the executive writes to and reads from.

THE VISION

A Second Brain for the Executive

V3.0 turns scattered operational history into a coherent **second brain** — the executive writes to it as it works and reads from it before it acts. Mem0 Cloud powers the store; git + markdown remains canonical.



Searchable

Query by goal, capability, incident pattern, or date range



Trustworthy

Accurate enough to plan against — confidence scoring per memory



Linkable

Goal → steps → contracts → outcomes → retros, all connected



Agent-Useful

Not just for humans — workers get context via memory packs

ARCHITECTURE

Mem0 Cloud hosts the knowledge store (api.mem0.ai). **Git + markdown** remains the canonical source of truth — mem0 is a derived, retrieval-optimized projection. If they disagree, **the repo wins**. The harvester produces high-signal memories only, not raw chat history.

4 Locked Pillars

1 Executive-Tier Only

Workers **never** call mem0, never hold the API key, never get the MCP server. The second brain is owned and accessed solely by the executive agent. Workers are *fed* relevant context via memory packs.

2 Git Is Source of Truth

Markdown in the private repo remains canonical. Mem0 stores **derived knowledge**, not authored content. If mem0 and the repo disagree, the repo wins. Mem0 can always be rebuilt from the repo.

3 Skills-First, Not a TS Module

No **src/agent/mem0/** data layer. The contract lives in markdown SKILL.md files. Deterministic helpers (schema validation, mem0 SDK calls) live inside each skill's **references/** folder — bundled with the skill, invoked via Bash.

4 One Writer

Only the **harvester skill**, run by the executive, writes to mem0. Direct **client.add()** from any worker or standalone script is an anti-pattern. Two agentic skills: memory-harvester (sole writer) and memory-reader (read-only).

Taxonomy v1.0.0 — 5 Memory Types



5 Lifecycle Hooks (A–E)

Wired into the executive loop, each gated by `V3_MEMORY_ENABLED` + per-hook flag. Staged rollout: one hook per week.

OFF

Hook A

Pre-Work Selection

Read — Consult memory before selecting work.

Currently audit-only until synthesis actually biases work selection. ~8-14 retrievals per consult.

ON

Hook B

Pre-Spawn Pack

Read → **Worker** — Search mem0 and bake a $\leq 2K$ -token memory pack into the worker's CLAUDE.md.

Workers consume static markdown.

ON

Hook C

Post-Run Harvest

Write — Classify outcome, write episodic memory after each run. Stage 1 of rollout. Lowest risk: writes only, no read decisions.

ON

Hook D

Failure Diagnosis

Read + Write — Surface prior failures with similar signals before diagnosis runs. Conditional write if a pattern is detected.

ON

Hook E

Post-Retro Harvest

Write — Write reflective memories from retro docs.

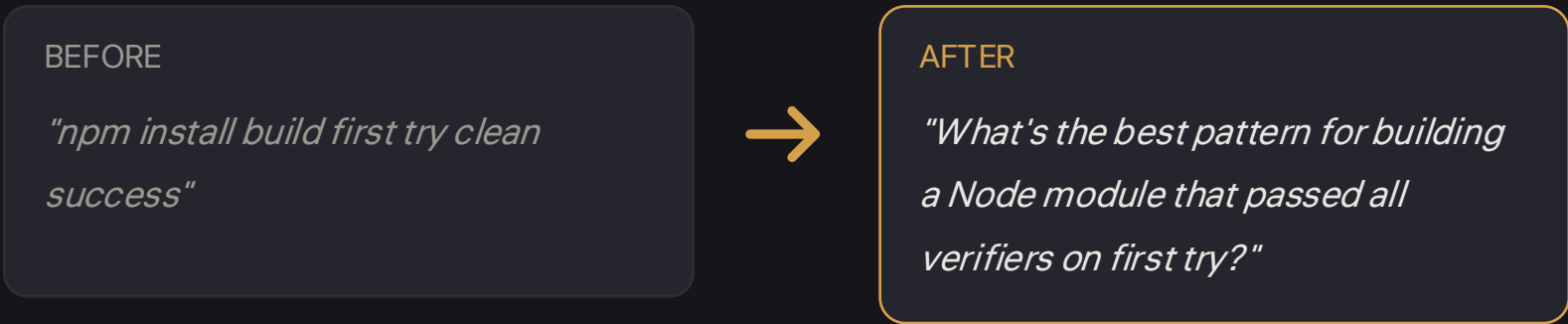
Highest judgment requirement. Saved for last so playbook is refined.

Agentic Retrieval + Memory Pack

THE SHIFT

The memory-reader was reframed from a fixed query checklist into an **agentic judgment task**:

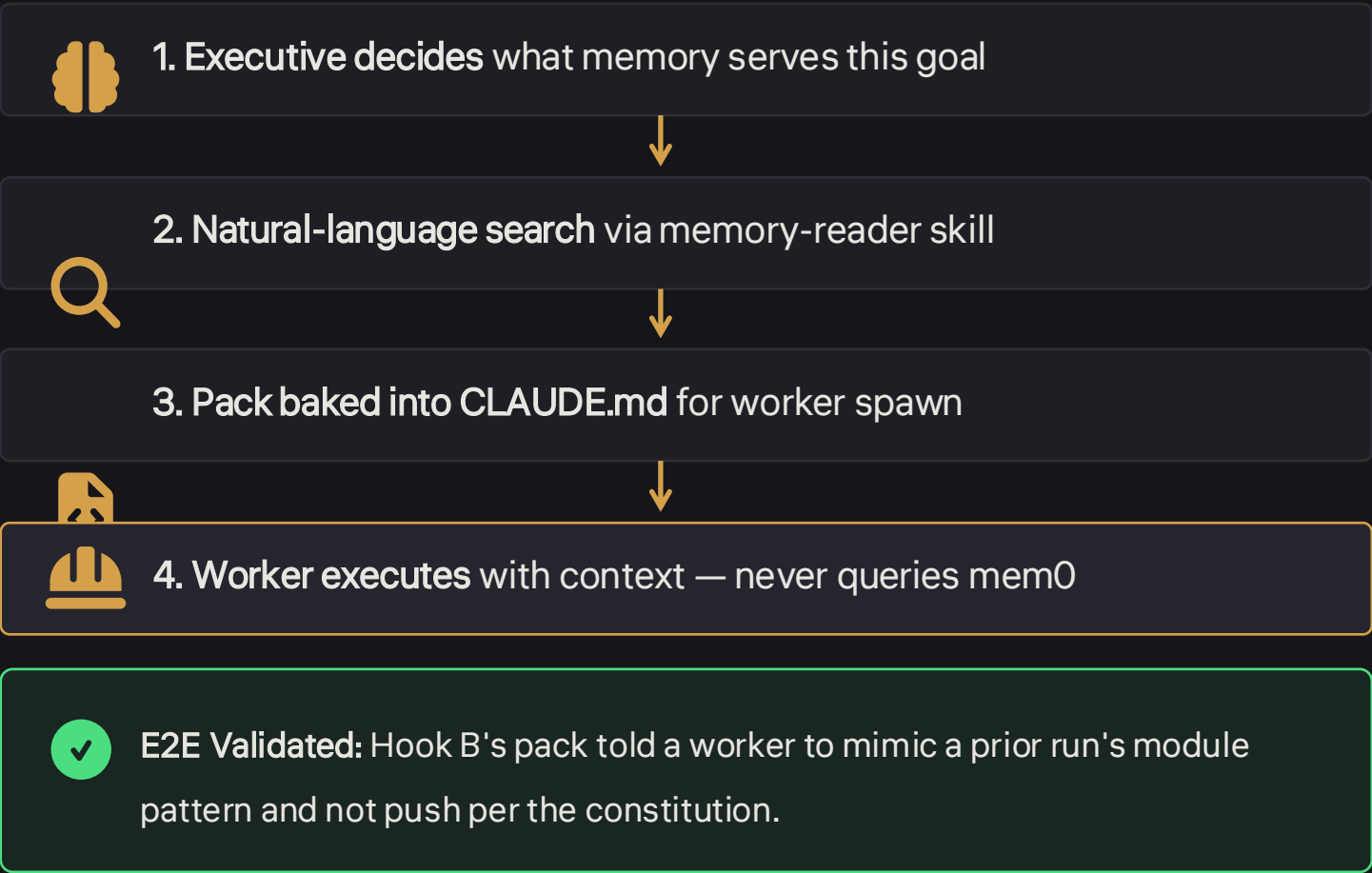
The executive decides which prior knowledge serves *this* goal and asks for it in **full natural-language sentences** — not short keyword bags.



3-8
refined queries per turn

≤2K
token memory pack cap

MEMORY PACK FLOW



INTEGRATION

Corpus Backfill + The Unified CLI

~134

memories migrated

4

harvest lanes

22

test memories purged

PASS

validator verdict

HARVEST LANES

Lane A — Principles

Constitution, hosting decision,
implementation plan → principle-type
memories

Lane B — Retros

Retro docs, learning logs → reflective +
semantic memories

Lane C — Capabilities

Capability registries → semantic
snapshots with expiry

Lane D — Episodic

Completed project bundles → episodic
outcome memories

UNIFIED CLI

The `./bin/mem0` CLI is the single deterministic surface the executive drives via Bash. Bakes in every
production gotcha:

Filter shape handling ▪ Snake/camel casing ▪ Auth ▪ Async-write event polling ▪ Schema validation

Daily snapshot

PM2 cron at 04:00 backs up the full mem0 store to JSON for
disaster recovery.

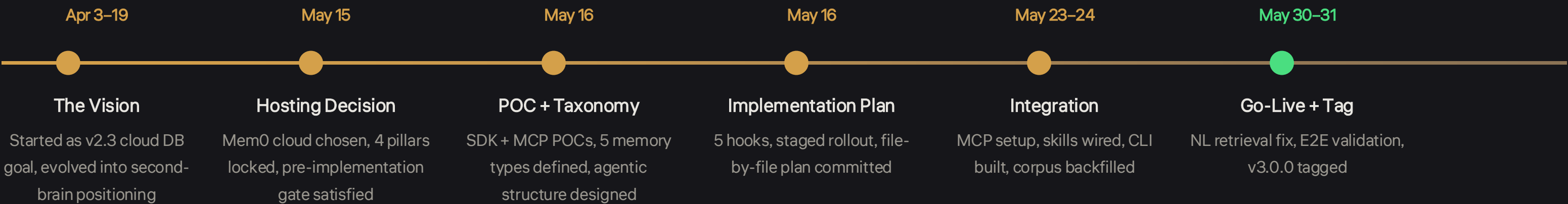
03

How We Did It

An 8-week journey from vision to live system. POC, taxonomy, skills, integration, go-live.

TIMELINE

The 8-Week Build Journey



KEY MILESTONES

**Two POCs in One Day**

SDK POC (graph-poc) + MCP POC (mcp-poc) both validated mem0 v3 async writes

**Taxonomy Defined**

5 types, 4-field ID convention, classify.ts validator — all in one session

**The Silent Bug**

Empty-string V3_MEM0_COHORT was silently failing every write — root-caused and fixed on go-live day

⏸

We have zero data on whether the playbooks produce useful memories; staging gives an audit window between each addition. — Implementation Plan, May 16

Changelog Highlights

ADDED

- mem0 second brain** — executive-tier cloud knowledge store
- memory-harvester skill** — sole writer, classifies + validates
- memory-reader skill** — read-only, NL search + pack builder
- ./bin/mem0 CLI** — unified deterministic surface
- 5 lifecycle hooks** (A–E) wired into exec loop
- Memory Pack injection** into worker CLAUDE.md
- Taxonomy v1.0.0** — 5 types, 4-field IDs
- Corpus backfill** — ~134 memories migrated
- Daily snapshot** — 04:00 DR backup cron

CHANGED

- Retrieval is agentic** — reframed from fixed query checklist to NL judgment task
- ai-docs reorganized** — chronologically by git-creation date, sequence is the point
- Skills decoupled** — from ai-docs, self-contained with inline references

FIXED

- Silent harvest failure** — empty-string V3_MEM0_COHORT rejected by validator, hook logged "ok"
- Hook observability** — now logs tool-name histogram + 400-char result tail
- Hook A idle burn** — fired every 30s idle iteration, now guarded by work queue

The Second Brain Is Live

The full read → pack → execute → write loop is validated end-to-end.

ROADMAP FORWARD



Hook A Activation

When synthesis biases selection



V3.1 Observability

Dashboard on trustworthy memory



Multi-Agent (V4.0)

Parallel agents, partitioned scope



Self-Hosted Path

mem0 OSS migration ready

AUTONOMOUS

SELF-IMPROVING

CONSTITUTION-GOVERNED